

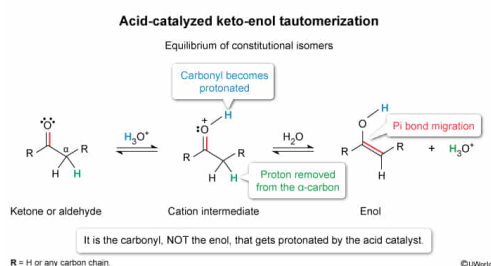
Which of the following statements does NOT correctly describe a keto-enol tautomerism?

- ☐ A. The reaction results in a pi bond migration, forming a C=C double bond. [10%]
- ✓ ☒ B. The reaction can be acid-catalyzed by protonating an enol oxygen atom. [39%]
- ☐ C. The reaction involves removal of a proton from an α -carbon. [21%]
- ☐ D. The reaction results in an equilibrium mixture of keto and enol tautomers, which are constitutional isomers. [28%]

Correct

40% answered correctly

Explanation:



Keto-enol tautomerization is a type of isomerization that involves a **proton removal** from the α -carbon of a **ketone or aldehyde (Choice C)** and addition of a proton to the carbonyl (C=O) oxygen atom. Additionally, a **pi bond** migrates from the C=O double bond to form a C=C double bond between the carbonyl carbon and the α -carbon (**Choice A**). The effective proton transfer and **pi bond migration** result in the formation of an **enol**. The ketone (or aldehyde) and enol forms of the molecule are called tautomers and exist in equilibrium. Because the tautomers have the same molecular formula and their structures only differ in the atom connectivity, they are classified as **constitutional isomers (Choice D)**.

Keto-enol tautomerization can be **acid catalyzed** when an acid **protonates** the carbonyl oxygen in the ketone (or aldehyde), resulting in a cation intermediate. Then, the α -carbon is deprotonated, and the pi bond migrates, forming an enol. Additionally, keto-enol tautomerizations can be **base catalyzed**.

The question asks which statement does NOT describe keto-enol tautomerization. Acid-catalyzed tautomerism does *not* involve protonation of the enol oxygen atom; instead, it involves protonation of the **carbonyl** oxygen atom.

Educational objective:

Keto-enol tautomerization is a base- or acid-catalyzed isomerization that converts a ketone or aldehyde into an enol. Tautomerism involves a proton transfer from the α -carbon of a ketone or aldehyde to the carbonyl oxygen atom along with pi bond migration. The keto and enol tautomers are in equilibrium and are classified as constitutional isomers.

Time spent: 0 sec

Subject:
Organic Chemistry

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System:
Functional Groups and Their Reactions